

BMATS101 – MATHEMATICS-I (CSE)

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MODULE 3

ORDINARY DIFFERENTIAL EQUATIONS (FIRST ORDER)

BASIC CONCEPTS, FORMATION & FIRST-ORDER EQUATIONS

1. Introduction to Differential Equations

A **differential equation** is an equation involving:

- An **unknown function**
- One or more of its **derivatives**

Differential equations are used to mathematically model **real-world phenomena** in engineering and computer science such as:

- Growth and decay of data
- Heat conduction
- Electrical circuits
- Motion of particles
- Population models

2. Ordinary Differential Equation (ODE)

2.1 Definition

A differential equation involving **ordinary derivatives** with respect to a **single independent variable** is called an **Ordinary Differential Equation**.

Example:

$$\frac{dy}{dx} = x^2$$

3. Order and Degree of a Differential Equation

3.1 Order

The **order** of a differential equation is the **highest order derivative** present.

Examples:

1. $\frac{dy}{dx} = x \rightarrow \text{Order} = 1$
2. $\frac{d^2y}{dx^2} + y = 0 \rightarrow \text{Order} = 2$

3.2 Degree

The **degree** of a differential equation is the **power of the highest order derivative**, after the equation is made free from radicals and fractions.

Example:

$$\left(\frac{dy}{dx}\right)^2 + y = 0$$

$\rightarrow \text{Order} = 1$

$\rightarrow \text{Degree} = 2$

Degree is defined **only when the equation is polynomial in derivatives**.

4. Formation of Differential Equations

Differential equations are formed by **eliminating arbitrary constants** from given relations.

Example 1

Form a differential equation from:

$$y = Ax + B$$

Solution

Differentiate w.r.t. x :

$$\frac{dy}{dx} = A$$

Differentiate again:

$$\frac{d^2y}{dx^2} = 0$$

This is the required differential equation.

Example 2

Form a differential equation from:

$$y = Ae^x + Be^{-x}$$

Differentiate:

$$\frac{dy}{dx} = Ae^x - Be^{-x}$$

Differentiate again:

$$\frac{d^2y}{dx^2} = Ae^x + Be^{-x}$$

Since:

$$y = \frac{d^2y}{dx^2}$$

The required differential equation is:

$$\boxed{\frac{d^2y}{dx^2} - y = 0}$$

5. General and Particular Solutions

5.1 General Solution

A solution of a differential equation which **contains arbitrary constants** is called the **general solution**.

Example:

$$y = Cx^2$$

5.2 Particular Solution

A solution obtained by assigning specific values to the constants using given conditions is called a **particular solution**.

6. First-Order Differential Equations

A differential equation whose **order is one** is called a **first-order differential equation**.

General form:

$$\frac{dy}{dx} = f(x, y)$$

7. Types of First-Order ODEs

In Module-3, VTU syllabus includes:

1. Linear differential equations
2. Bernoulli's differential equations
3. Exact differential equations

4. Reducible to exact equations

5. Non-linear equations

- Solvable for p
- Clairaut's equation

8. Linear Differential Equations

8.1 Definition

A first-order linear differential equation is of the form:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where:

- $P(x), Q(x)$ are functions of x

8.2 Integrating Factor (I.F.)

The **integrating factor** is defined as:

$$\text{I.F.} = e^{\int P(x) dx}$$

8.3 Solution of Linear Differential Equation

Multiply the equation by I.F.:

$$\frac{d}{dx} [y \cdot \text{I.F.}] = Q(x) \cdot \text{I.F.}$$

Integrate both sides:

$$y \cdot \text{I.F.} = \int Q(x) \cdot \text{I.F.} dx + C$$

9. Solved Problems – Linear ODE

Example 3

Solve:

$$\frac{dy}{dx} + y = e^x$$

Solution

Comparing with standard form:

$$P(x) = 1, Q(x) = e^x$$

Integrating factor:

$$\text{I.F.} = e^{\int 1 dx} = e^x$$

Multiply throughout by I.F.:

$$e^x \frac{dy}{dx} + ye^x = e^{2x}$$

Left side becomes:

$$\frac{d}{dx}(ye^x)$$

Integrate:

$$ye^x = \int e^{2x} dx = \frac{e^{2x}}{2} + C$$

Hence:

$$y = \frac{e^x}{2} + Ce^{-x}$$

12. Practice Problems

1. Solve: $\frac{dy}{dx} - y = x$
2. Solve: $\frac{dy}{dx} + 2y = \sin x$
3. Solve: $\frac{dy}{dx} + y \tan x = \sec x$

BERNOULLI, EXACT & REDUCIBLE-TO-EXACT DIFFERENTIAL EQUATIONS

1. Bernoulli's Differential Equation

1.1 Definition

A first-order differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

where $n \neq 0, 1$, is called **Bernoulli's differential equation**.

1.2 Reduction to Linear Form

Divide the equation by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = Q(x)$$

Let:

$$v = y^{1-n}$$

Then:

$$\frac{dv}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

The equation reduces to a **linear differential equation in v** .

1.3 Solved Problem (Bernoulli)

Example 1

Solve:

$$\frac{dy}{dx} + y = xy^2$$

Solution

Compare with standard form:

$$P(x) = 1, Q(x) = x, n = 2$$

Divide by y^2 :

$$y^{-2} \frac{dy}{dx} + y^{-1} = x$$

Let:

$$v = y^{-1} \Rightarrow \frac{dv}{dx} = -y^{-2} \frac{dy}{dx}$$

Hence:

$$-\frac{dv}{dx} + v = x$$

or

$$\frac{dv}{dx} - v = -x$$

This is linear.

Integrating factor:

$$\text{I.F.} = e^{\int (-1)dx} = e^{-x}$$

Solve:

$$ve^{-x} = \int (-xe^{-x})dx + C$$

Back-substitute $v = \frac{1}{y}$ to get final answer.

2. Exact Differential Equations

2.1 Definition

A differential equation of the form

$$M(x, y) dx + N(x, y) dy = 0$$

is said to be **exact** if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

2.2 Solution of Exact Equation

If exact, then the solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy = C$$

2.3 Solved Problem (Exact Equation)

Example 2

Solve:

$$(2xy + y^2) dx + (x^2 + 2xy) dy = 0$$

Solution

Let:

$$M = 2xy + y^2, N = x^2 + 2xy$$

Compute partial derivatives:

$$\frac{\partial M}{\partial y} = 2x + 2y$$

$$\frac{\partial N}{\partial x} = 2x + 2y$$

Since equal, equation is **exact**.

Integrate M w.r.t. x :

$$\int (2xy + y^2) dx = x^2 y + xy^2 + \phi(y)$$

Differentiate w.r.t. y and equate to N :

$$\Rightarrow \phi'(y) = 0$$

Hence solution:

$$x^2 y + xy^2 = C$$

3. Differential Equations Reducible to Exact Form

Some equations are **not exact directly**, but become exact after multiplication by an **Integrating Factor (I.F.)**.

3.1 Integrating Factor Rules (VTU IMPORTANT)

Case 1

If:

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$$

is a function of x only, then:

$$\text{I.F.} = e^{\int f(x) dx}$$

Case 2

If:

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M}$$

is a function of y only, then:

$$\text{I.F.} = e^{\int f(y) dy}$$

3.2 Solved Problem (Reducible to Exact)

Example 3

Solve:

$$(y + 2x) dx + x dy = 0$$

Solution

Let:

$$M = y + 2x, N = x$$
$$\frac{\partial M}{\partial y} = 1, \frac{\partial N}{\partial x} = 1$$

Here:

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 0$$

Hence equation is already **exact**.

Integrate:

$$\int (y + 2x) dx = xy + x^2$$

Thus solution:

$$xy + x^2 = C$$

4. Practice Problems

1. Solve: $(y - x) dx + (x - y) dy = 0$
2. Solve: $(x^2 + y) dx + x dy = 0$
3. Solve Bernoulli equation:

$$\frac{dy}{dx} + y = xy^3$$

ORTHOGONAL TRAJECTORIES, L-R & C-R CIRCUITS, APPLICATIONS

1. Orthogonal Trajectories

1.1 Definition

Two families of curves are said to be **orthogonal trajectories** of each other if **each curve of one family cuts every curve of the other family at right angles (90°)**.

Mathematically, if the slopes of two curves at the point of intersection are m_1 and m_2 , then:

$$m_1 m_2 = -1$$

2. Method to Find Orthogonal Trajectories

Step-by-step Procedure

1. Write the given family of curves.
2. Differentiate w.r.t. x to obtain a differential equation.

3. Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$.
4. Solve the resulting differential equation.
5. The solution represents the **orthogonal trajectories**.

3. Solved Problems – Orthogonal Trajectories

Example 1

Find the orthogonal trajectories of the family:

$$y = Cx$$

Solution

Differentiate w.r.t. x :

$$\frac{dy}{dx} = C$$

Since $C = \frac{y}{x}$,

$$\frac{dy}{dx} = \frac{y}{x}$$

Replace:

$$\begin{aligned}\frac{dy}{dx} &\rightarrow -\frac{dx}{dy} \\ -\frac{dx}{dy} &= \frac{y}{x} \\ \Rightarrow x \, dx + y \, dy &= 0\end{aligned}$$

Integrate:

$$\frac{x^2}{2} + \frac{y^2}{2} = C$$

Answer

$$\boxed{x^2 + y^2 = C}$$

Example 2 (Very Important)

Find orthogonal trajectories of:

$$x^2 + y^2 = C$$

Solution

Differentiate:

$$2x + 2y \frac{dy}{dx} = 0$$
$$\frac{dy}{dx} = -\frac{x}{y}$$

Replace:

$$-\frac{dx}{dy} = -\frac{x}{y} \Rightarrow \frac{dx}{dy} = \frac{x}{y}$$
$$\frac{dx}{x} = \frac{dy}{y}$$

Integrate:

$$\ln x = \ln y + \ln C$$
$$\Rightarrow x = Cy$$

4. Applications of Orthogonal Trajectories

- Streamlines and equipotential lines
- Electric field lines
- Heat flow
- Fluid flow modeling

5. Application of ODEs – Growth and Decay

5.1 Law of Natural Growth/Decay

The rate of change of a quantity is **proportional to the quantity itself**.

$$\frac{dy}{dt} \propto y$$
$$\boxed{\frac{dy}{dt} = ky}$$

Solution

$$\frac{dy}{y} = k dt$$

Integrate:

$$\ln y = kt + C$$
$$\boxed{y = Ae^{kt}}$$

Where:

- $k > 0 \rightarrow$ growth
- $k < 0 \rightarrow$ decay

Example 3 (Growth)

If population doubles in 10 years, find growth law.

Solution

$$y = Ae^{kt}$$

When $t = 10$, $y = 2A$:

$$2A = Ae^{10k} \Rightarrow k = \frac{\ln 2}{10}$$

Example 4 (Decay)

Find decay law of radioactive substance.

$$y = Ae^{-kt}$$

6. Application of ODEs – L–R Circuit

6.1 L–R Circuit Equation

For an L–R circuit:

$$L \frac{di}{dt} + Ri = E$$

Where:

- L = inductance
- R = resistance
- E = electromotive force
- i = current

Solution

Divide by L :

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$$

This is a **linear differential equation**.

Integrating factor:

$$\text{I.F.} = e^{\frac{R}{L}t}$$

Final solution:

$$i = \frac{E}{R} + C e^{-\frac{R}{L}t}$$

7. Application of ODEs – C–R Circuit

7.1 C–R Circuit Equation

For a C–R circuit:

$$RC \frac{dv}{dt} + v = E$$

Where:

- R = resistance
- C = capacitance
- v = voltage

Solution

Divide by RC :

$$\frac{dv}{dt} + \frac{1}{RC}v = \frac{E}{RC}$$

Integrating factor:

$$\text{I.F.} = e^{\frac{t}{RC}}$$

Solution:

$$v = E + C e^{-\frac{t}{RC}}$$

8. Practice Problems

1. Find orthogonal trajectories of $y^2 = Cx$
2. Solve growth equation when quantity triples in 5 years
3. Solve L–R equation for given initial current
4. Solve C–R equation for voltage across capacitor

NON-LINEAR DIFFERENTIAL EQUATIONS

(Solvable for p , Clairaut's Equation & Singular Solutions)

1. Introduction to Non-Linear Differential Equations

A **non-linear differential equation** is a differential equation which is **not linear** in the dependent variable or its derivatives.

These equations arise in:

- Mechanical systems
- Control systems
- Optimization problems
- Engineering modelling

In VTU Module-3, the following **non-linear equations** are included:

- Equations solvable for p
- Clairaut's equations
- Singular solutions

2. Differential Equations Solvable for p

Let

$$p = \frac{dy}{dx}$$

If a differential equation can be expressed in the form:

$$y = xp + f(p)$$

or

$$x = yp + f(p)$$

then it is called an equation **solvable for p** .

3. Method of Solution (Solvable for p)

Given:

$$y = xp + f(p)$$

Differentiate both sides with respect to x :

$$\frac{dy}{dx} = p + x \frac{dp}{dx} + f'(p) \frac{dp}{dx}$$

Since $\frac{dy}{dx} = p$,

$$\begin{aligned} p &= p + (x + f'(p)) \frac{dp}{dx} \\ \Rightarrow (x + f'(p)) \frac{dp}{dx} &= 0 \end{aligned}$$

This gives two cases:

Case 1:

$$\frac{dp}{dx} = 0$$

Case 2:

$$x + f'(p) = 0$$

4. Solved Problem – Solvable for p

Example 1

Solve:

$$y = xp + p^2$$

Solution:

Differentiate:

$$\frac{dy}{dx} = p + x \frac{dp}{dx} + 2p \frac{dp}{dx}$$

Since $\frac{dy}{dx} = p$,

$$p = p + (x + 2p) \frac{dp}{dx}$$

$$(x + 2p) \frac{dp}{dx} = 0$$

Case 1:

$$\frac{dp}{dx} = 0 \Rightarrow p = C$$

Substitute in original equation:

$$y = Cx + C^2$$

This is the **general solution**.

Case 2:

$$x + 2p = 0 \Rightarrow p = -\frac{x}{2}$$

Substitute:

$$y = x \left(-\frac{x}{2} \right) + \left(-\frac{x}{2} \right)^2$$

$$y = -\frac{x^2}{2} + \frac{x^2}{4}$$

$$y = -\frac{x^2}{4}$$

This is the **singular solution**.

5. Clairaut's Differential Equation

5.1 Definition

A differential equation of the form:

$$y = xp + f(p)$$

where $p = \frac{dy}{dx}$,

is called **Clairaut's equation**.

5.2 Method of Solution

Step 1: Differentiate with respect to x

$$\frac{dy}{dx} = p + x \frac{dp}{dx} + f'(p) \frac{dp}{dx}$$

Step 2: Since $\frac{dy}{dx} = p$,

$$(x + f'(p)) \frac{dp}{dx} = 0$$

Step 3: Solve the two cases to obtain:

- General solution
- Singular solution

6. Solved Problems – Clairaut's Equation

Example 2

Solve:

$$y = xp + p^2$$

Solution:

This is a Clairaut's equation.

General solution:

$$p = C \Rightarrow y = Cx + C^2$$

Singular solution:

$$x + 2p = 0 \Rightarrow p = -\frac{x}{2}$$

Substitute:

$$y = -\frac{x^2}{4}$$

7. Another Solved Example

Example 3

Solve:

$$y = xp + \sin p$$

Solution:

Differentiate:

$$\frac{dy}{dx} = p + x \frac{dp}{dx} + \cos p \frac{dp}{dx}$$

$$(x + \cos p) \frac{dp}{dx} = 0$$

Case 1:

$$p = C \Rightarrow y = Cx + \sin C$$

(General solution)

Case 2:

$$x + \cos p = 0 \Rightarrow x = -\cos p$$

This gives the **singular solution** in parametric form.

8. Singular Solution

8.1 Definition

A **singular solution** is a solution:

- Which is **not obtained** by putting specific values of constants in the general solution
- Which represents the **envelope** of the family of curves defined by the general solution

9. Geometrical Interpretation

- General solution $\rightarrow$ family of straight lines
- Singular solution $\rightarrow$ envelope touching all curves

This is often asked as a **theory question in VTU**.

10. Practice Problems

1. Solve $y = xp + 3p^2$
2. Solve $y = xp + e^p$
3. Find singular solution of $y = xp - p^2$